NHH

SCHOOL EXAMINATION

BAN404

Spring, 2024

Date: 27.05.2024

Time: 9.00-15.00

Number of hours: 6

An invigilator can contact course responsible by phone: 47078891

SUPPORT MATERIALS PERMITTED DURING THE EXAMINATION:

Calculator

All written support material permitted (category III)

Number of pages, including front page: 4

Number of attachments: 1, The file “airline.csv”.

Instructions and recommendations

General recommendations: Be pragmatic if you run into difficulties. You need to find models that can answer the questions asked. Don´t let the perfect model stand in the way for a good one! Also, plan your work so that you do not use all time to solve one task.

R coding can be done in different ways, e.g. using base R or the tidyverse approach. Given a correct and clearly written result there is no preference for any particular approach.

Throughout the exam you will work with training and test data and you will have to make appropriate choices on when to use one or the other. In cases when you think there are more than one valid choice on working with the training, test or the full dataset, explain your choice!

Tasks

1. (50 points) This task is about some of the methodological topics treated in the course.

   (a) (10 points) Explain what the following R-function is doing.

Solution:

```
f <- function(x0,x,y,K=3)
{
  d <- abs(x-x0) # Distance between x-vals and x0
  o <- order(d)[1:K] # Index for K smallest d
  xl <- x[o] # Choose x's closest to x0
  yl <- y[o] # Choose corresponding y's
  xydata <- data.frame(xl=xl,yl=yl) # Store as df
  reg <- lm(yl~xl,data=xydata) # Regress yl on xl
  ypred <- predict(reg,newdata=data.frame(xl=x0,yl=1))
  # Predict y for x0 based on regression
  return(ypred)
}
```

See comments in R code. The method resambles KNN but instead of taking the average of the y's corresponding to the x-values closest to x0, this method run a regression of the same values and predicts based on it.

   (b) (10 points) Consider the following small dataset

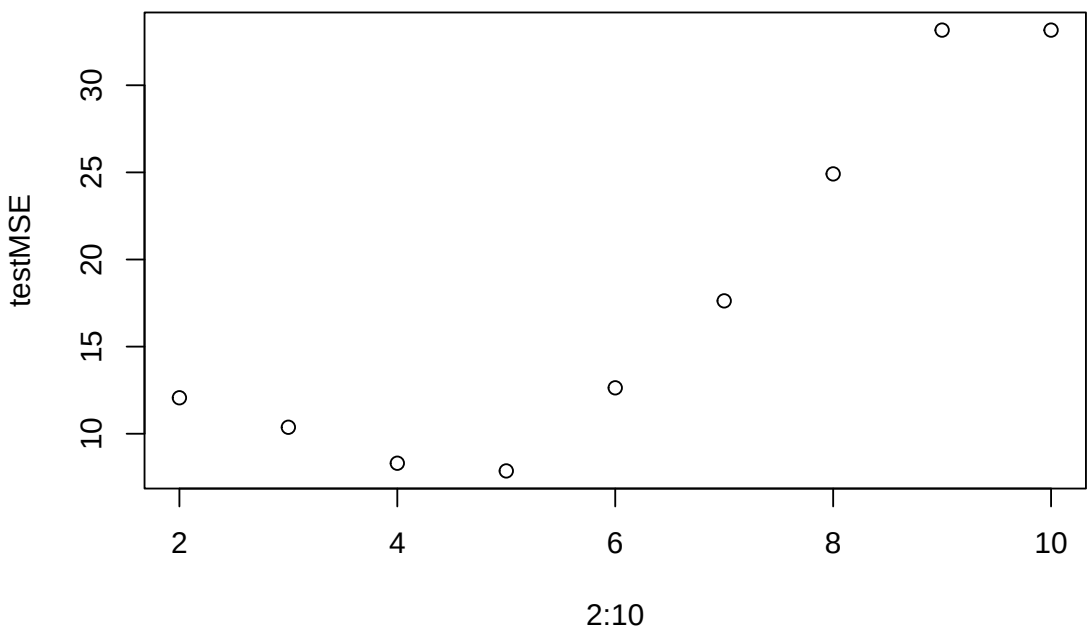
x	1	2	3	4	5	6	7	8	9	10
y	5.26	9.13	11.17	15.64	25.32	25.55	41.39	48.17	58.65	68.24

Use leave-one-out cross-validation to determine the optimal K in the function `f` for this dataset.

Solution:

```r
# LOO determination of K
loo <- function(K,x,y)
{
  n <- length(x)
  ypred <- matrix(0,n,1)
  for(i in 1:n)
  {
    ypred[i] <- f(x[i],x[-i],y[-i],K=K)
  }
  testMSE <- mean((y-ypred)^2)
  return(testMSE)
}

testMSE <- matrix(0,9,1)
for(k in 2:10) testMSE[k-1] <- loo(k,x,y)
plot(2:10,testMSE)
```

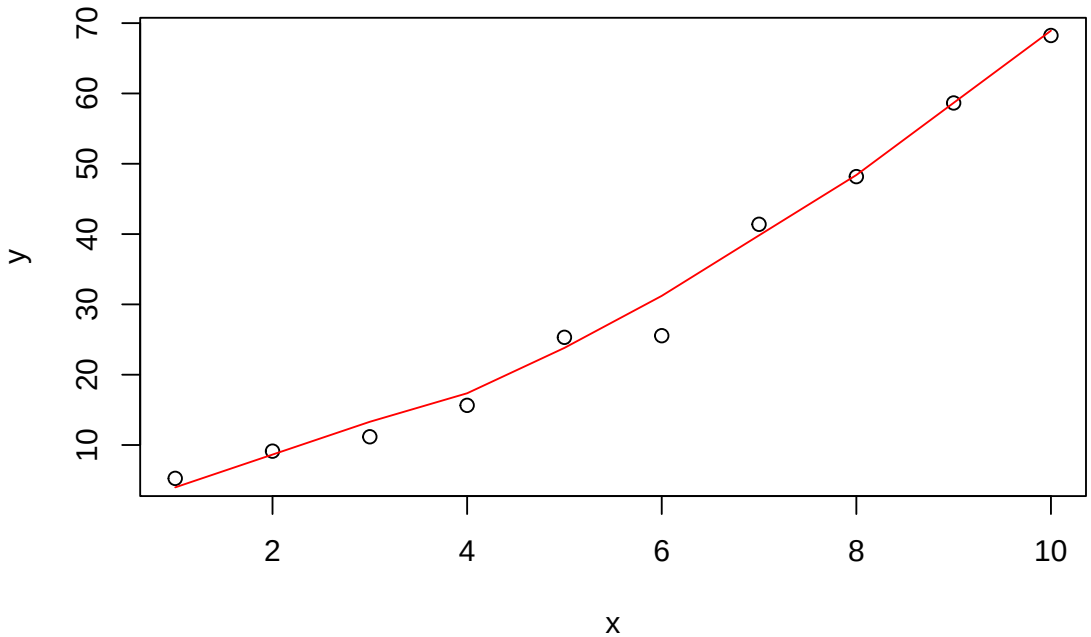


As can be seen the minimum testMSE is found for $K = 5$

(c) (10 points) Plot y against x and add a line with the predictions based on the optimal K.

Solution:

```r
ypred <- matrix(0,n,1)
for(i in 1:n) ypred[i] <- f(x[i],x,y,K=5)
plot(x,y)
lines(x,ypred,col="red")
```



(d) (10 points) In the function `f`, there is only one predictor. One way to allow for more than one predictor is to compute `d` in the `f`-function in (a) differently. Explain how such a modification can be done; examplify with a case with two predictors.

Solution:

`d` is then computed as the Euclidian distance in p-dimension, where p is the number of predictors. Let $\boldsymbol{x} = (x_1, x_2)'$ be a training observation and $\boldsymbol{x}_0 = (x_{01}, x_{02})'$. Then

$$d = \sqrt{(x_1 - x_{01})^2 + (x_2 - x_{02})^2}$$

(e) (10 points) Another way to allow for several predictors is to use backfitting. Explain how you would implement backfitting for this method (you do not need to do it).

Solution:

I examplify with the case of two predictors. We start by fitting the model of y on x_1 using the function `f` in (a). The residuals from this, $e_1 = y - \hat{f}(x_1)$ is then

used as the y-variable in the next iteration with x_1 substituted with x_2. The new residuals $e_2 = y - \hat{f}(x_2)$ is subsequently used in the next iteration where, again, x_1 is used as the predictor. When this is repeated a few times the estimated functions are not changing anymore and we have estimates both for $f(x_1)$ and $f(x_2)$.

2. (50 points) In this task you shall analyze the dataset in the file `airline.csv`. This data is collected by Jana (n.d.). You can read it into R with the `read.csv` function (or `read_csv` if you prefer the tidyverse approach).

```
airline <- read.csv("airline.csv")
```

The dataset contains answers from a survey of customers of an airline. The customers are anonymous to the airline. You will help the airline with two tasks. First, you shall analyze **why some of the customers are unsatisfied**. Secondly, you shall help the airline to **predict if a new customer will be satisfied on the next flight**. The airline will use your prediction model to identify customers likely to be unsatisfied and put in place measures to reduce this likelihood. The variables in the data set are given in the table below. For the detailed customer satisfaction questions the answers is a number between 0 and 5 meaning where 0 = no answer, 1 = very unhappy,. . . , 5 = very happy.

Variable name	**Explanation**
satisfaction	Overall satisfaction of customer
Age	Age of customer
Arrival_Delay_in_Minutes	Delay of arrival
Baggage_handling	Customer's satisfaction of bagage handling
Checkin_service	Customer's satisfaction of checkin service
Class	Passenger class
Cleanliness	Customer's satisfaction of checkin service
Customer_Type	Loyal or disloyal customer
DA_time_convenient	Customer's view on departure/arrival time
Departure_Delay_in_Minutes	Departure delay
Ease_of_Online_booking	Customer's satisfaction of online booking
Flight_Distance	Distance of flight
Food_and_drink	Customer's satisfaction of foods and drinks
Gate_location	Customer's satisfaction of gate location
Gender	Gender
Inflight_entertainment	Customer's satisfaction of inflight entertainment
Inflight_wifi_service	Customer's satisfaction of inflight entertainment
Leg_room_service	Customer's satisfaction of leg room
On_board_service	Customer's satisfaction of onboard service
Online_boarding	Customer's satisfaction of online check-in
Online_support	Customer's satisfaction of online support
Seat_comfort	Customer's satisfaction of seat comfort
Type_of_Travel	Customer's type of travel, e.g. business

(a) (3 points) If necessary, recode categorical variables as factors and remove variables

that cannot be used in the analysis. Motivate your choices in words.

Solution:

I recode the variables with less than 8 categories as factor variables.

```
set.seed(123)
p <- ncol(airline)
n <- nrow(airline)
for(i in 1:p) if(length(table(airline[,i]))<=7)
  airline[,i] <- as.factor(airline[,i])
ind <- sample(1:n,size=floor(n/2))
training <- airline[ind,]
test <- airline[-ind,]
```

(b) (3 points) In tasks (a)-(g) the methods and models is used to answer the question "Why are some customers dissatisfied?". With this in mind, motivate your choice to evaluate the models (on all/training/test data).

Solution:

Even though the question was not explicitly to predict if a customer is satisfied, I have chosen to split the data into a training and test part and do all descriptive analysis and model fitting on the training data. This is because I had already read the questions later in the task when operational predictions was treated. I therefore wanted to be able to evaluate the extra prediction power gained by using the survey questions in the original data set. The analysis here will therefore be directly comparable with the results later. It is also a safe-guard for not using test data in specifying the model in the (h)-(l).

It would also be correct to argue to use the entire dataset in (a)-(g). The arguments could then be: The question "Why are some customers dissatisfied?" is not one about prediction so an in-sample analysis. A benefit compared to splitting the data in parts is then that we can use more data to infer information from the data.

(c) (3 points) Use descriptive statistics to find variables associated with `satisfaction`. First, explain which type of descriptive statistics (types of tables and figures) that you are using. Give examples of R code but do not show all code and output if you are doing the same thing many times. Summarize your results as text, mentioning the relevant numbers.

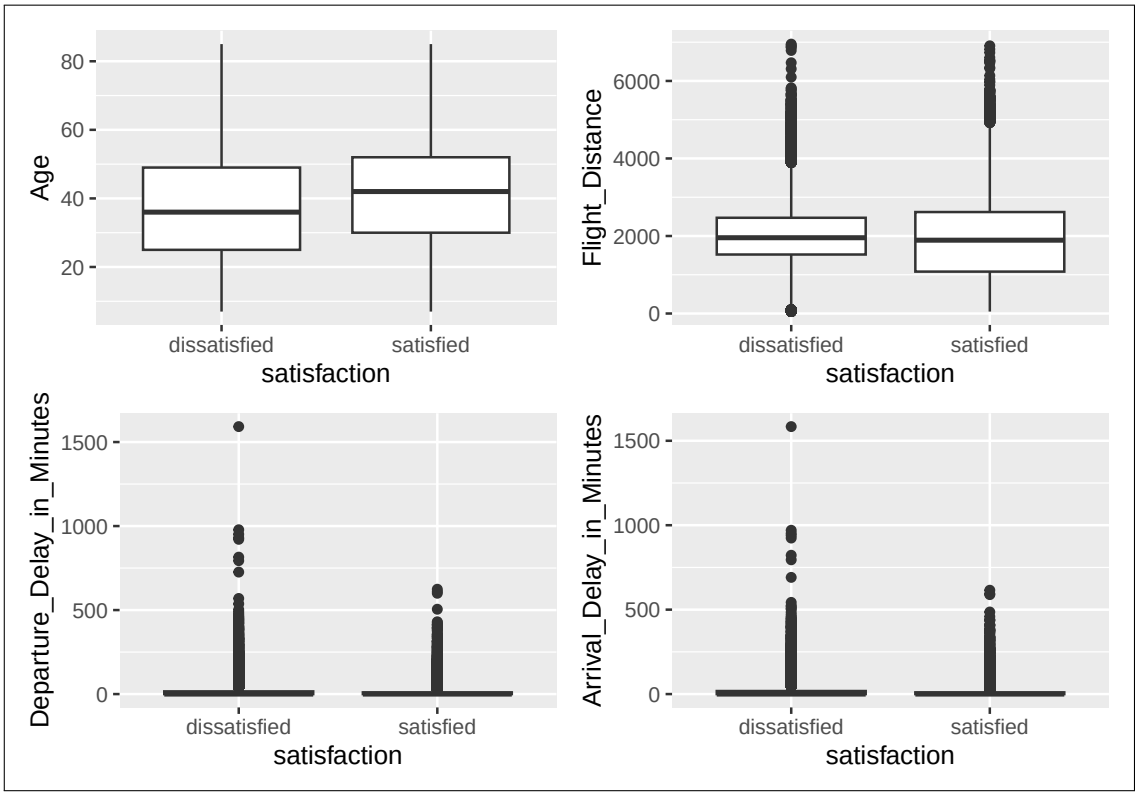
Solution:

The fraction of satisfied customers in the dataset is 54% which was obtained by the R-code `prop.table(table(airline$satisfaction))`. I then look at fractions, conditional on potential predictors, by first creating a function for this. I examplify by applying the function to the variable `Gender`. We can see that males are, on average, less content with the airline, than females are. This function is then applied to all the factor variables. The result are mostly as expected. Loyal customers are more satisfied than disloyal ones. Business travelers and business class travelers are more satisfied than other passenger groups. The more nuanced questions on satisfaction, such as the quality of onboard service, cleanliness etc are, as anticipated, positively associated with the `satisfaction` variable itself. This is interesting and can give insight into why a customer might not be satisfied.

```r
proptab <- function(x) round(100*prop.table(
  table(airline[,x],airline$satisfaction),margin=1))
proptab("Gender") # Females more satisfied
```

```
##
##          dissatisfied satisfied
##   Female           35        65
##   Male             56        44
```

As can be seen on the boxplots below for the satisfied, and dissatisfied, respectively, age is on average higher for the satisfied customers. No obvious pattern for flight distance. Flight distance and delays of departures and arrivals have no strong association with satisfaction.



(d) (5 points) Based on the results in the previous tasks, formulate a logistic regression model and use it to evaluate which variables are associated with satisfaction. Evaluate the model with one or more appropriate measures of model fit.

Solution:

I choose to evaluate the predictions with confusion tables and overall accurracy. A correct use of ROC/AUC also give full score. In a first attempt, with all predictors except `DA_time_convenient`, several coefficients is estimated as `NA`. A possible reason for this is that many of the predictors are highly dependent of each others (dissatisfaction of one type is often associated with other types). After removing also `Cleanliness` and `Online_boarding` the model is appropriately estimated. However, many coefficients have large p-values. That is, many variables are not statistically significant. I choose to remove those with the hope of getting a model which producing better predictions.

```r
logreg1 <- glm(satisfaction~.-DA_time_convenient-Cleanliness-
                 Online_boarding,data=training,family=binomial())
formula2 <- satisfaction~Gender+Customer_Type+Age+
  Type_of_Travel + Class + Flight_Distance +
  Food_and_drink + Inflight_entertainment +
  Online_support + Baggage_handling +
  Checkin_service + Arrival_Delay_in_Minutes
logreg2 <- glm(formula2,data=training,family=binomial())
```

A confusion matrix for this model is

```r
prob2 <- predict(logreg2,newdata=test,type="response")
pred2 <- prob2 > 0.5
conf_matrix <- table(test$satisfaction,pred2)
conf_matrix
```

```
##                pred2
##                 FALSE  TRUE
##   dissatisfied  9582  1738
##   satisfied     1912 11691
```

```r
prop.table(conf_matrix,margin=1)
```

```
##                pred2
##                     FALSE      TRUE
##   dissatisfied 0.8464664 0.1535336
##   satisfied    0.1405572 0.8594428
```

```r
overall_accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
overall_accuracy
```

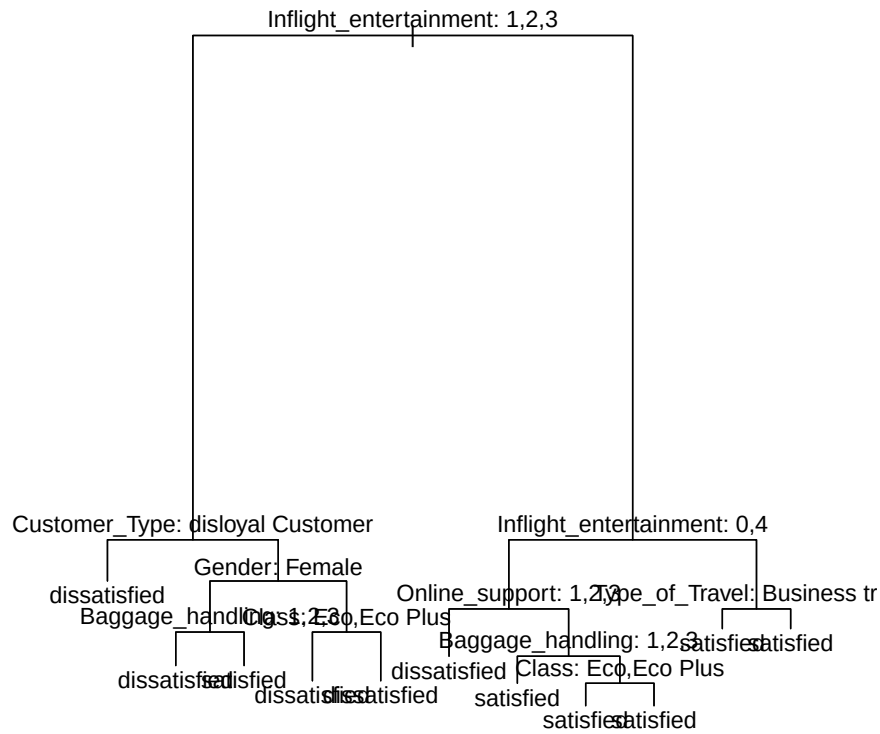
```
## [1] 0.8535489
```

(e) (5 points) Fit a classification tree, plot it and interpret it. Evaluate the model fit.

Solution:

Using the same variables as in the logistic regression we obtain

```r
library(tree)
tree1 <- tree(formula2,data=training)
```



As an example, a customer dissatisfied with in-flight entertainment, who is a loyal customer, is female and satisfied with the baggage handling is predicted to answer "satisfied" on the target question.

```
prob_tree <- predict(tree1,newdata=test)
pred_tree <- prob_tree[,2]>0.5
conf_matrix <- table(test$satisfaction,pred_tree)
conf_matrix
```

```
##                pred_tree
##                 FALSE  TRUE
##   dissatisfied   9296  2063
##   satisfied      2063 11578
```

```
prop.table(conf_matrix,margin=1)
```

```
##                pred_tree
##                     FALSE      TRUE
##   dissatisfied 0.8183819 0.1816181
##   satisfied    0.1512352 0.8487648
```

```
overall_accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
overall_accuracy
```

```
## [1] 0.83496
```

Somewhat worse predictions than for the logistic model.

(f) (5 points) Can you improve the predictions by using bagging or a random forest? Compare the model fit with the models in the previous tasks. Compute variable importance measure and interpret it.

Solution:

```
library(randomForest)
rf <- randomForest(formula2,data=training,na.action = na.omit)
pred_rf <- predict(rf,newdata=test)
conf_matrix <- table(test$satisfaction,pred_rf)
conf_matrix
```

```
##                pred_rf
##                 dissatisfied satisfied
##   dissatisfied        10259      1061
##   satisfied             992     12611
```

```
prop.table(conf_matrix,margin=1)
```

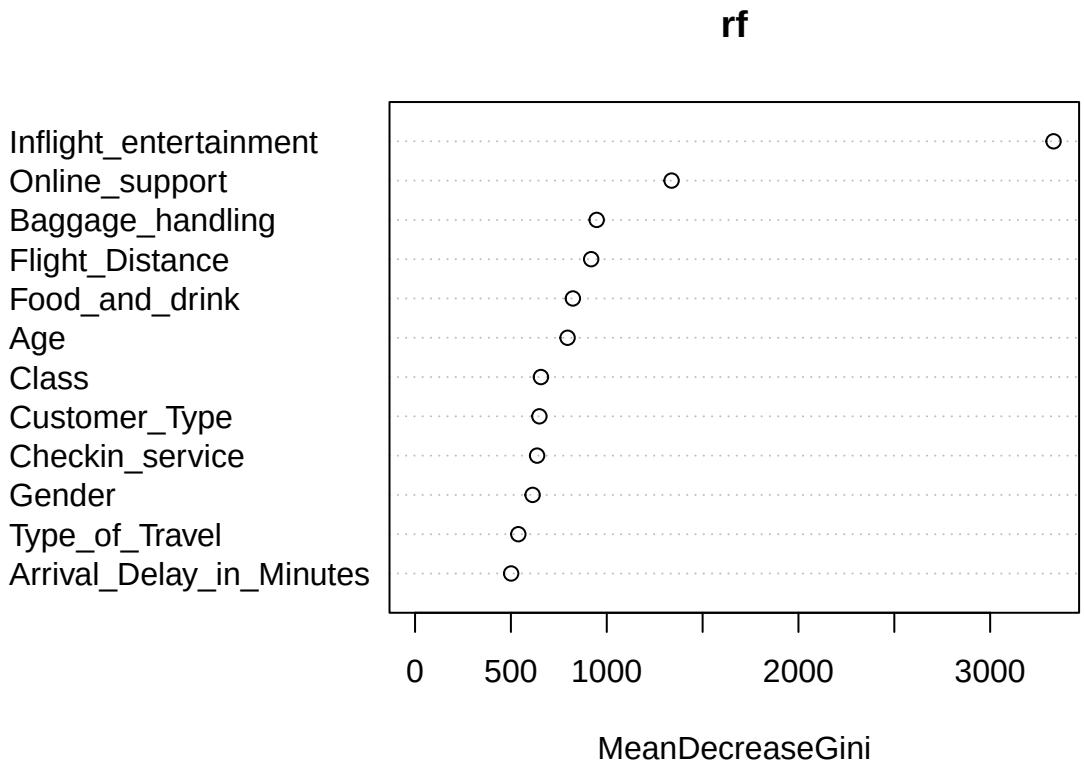
```
##                pred_rf
##                 dissatisfied  satisfied
##   dissatisfied   0.90627208 0.09372792
##   satisfied      0.07292509 0.92707491
```

```
overall_accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
overall_accuracy
```

```
## [1] 0.9176263
```

A significant improvement compared to previous methods.

```
varImpPlot(rf)
```



(g) (4 points) Based on the data analysis so far, what is your answer to the airline's question "Why are some customers dissatisfied?"

Solution:

All models so far indicates that knowledge of a customer's satisfaction of in-flight entertainment, online support and baggage handling are important factors in the customer's overall satisfaction.

(h) (4 points) You should now, and for the rest of this task, shift perspective. The airline also would like to know **how they can reduce the dissatisfaction of those customers**. Since the survey data is anonymous, a prediction model for who will potentially be unhappy about their flight must therefore account for the fact that all variables will not be available for a customer who have bought a ticket. State and explain your assumptions about which of the variables are observed for a customer who has bought a ticket.

Solution:

I assume that the variables stating the customer's satisfaction about certain aspects of their flight will be unknown to the airline at the time of the next

ticket purchase. I therefore do not use the variables described with the words "Customer's satisfaction of...", the delay variables and the gate location in the table in Table 2.

(i) (2 points) In tasks (h)-(l) the methods and models is used to answer the question "How can we predict the likelihood of a new customer to be dissatisfied?". With this in mind, motivate your choice to evaluate the predictions (on all/training/test data).

Solution:

To mimic the real-world situation of actually predicting new, unseen, customers it is essential to split the data up in training and test data.

(j) (5 points) Formulate a logistic model which can be used to predict the probability of a customer, who has already bought a ticket, will be unsatisfied if nothing is done to stop it. Evaluate the predictions in an appropriate way.

Solution:

```
formula_op <- satisfaction~Gender + Customer_Type +
  Age+Type_of_Travel+ Class + Flight_Distance
logreg_op <- glm(formula_op,data=training,family=binomial())
prob2_op <- predict(logreg_op,newdata=test,type="response")
pred2_op <- prob2_op > 0.5
conf_matrix <- table(test$satisfaction,pred2_op)
conf_matrix
```

```
##                pred2_op
##                 FALSE  TRUE
##   dissatisfied  7427  3932
##   satisfied     2064 11577
```

```
prop.table(conf_matrix,margin=1)
```

```
##                pred2_op
##                     FALSE      TRUE
##   dissatisfied 0.6538428 0.3461572
##   satisfied    0.1513086 0.8486914
```

```
overall_accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
overall_accuracy
```

```
## [1] 0.76016
```

(k) (7 points) Use a random forest to predict whether a customer will be dissatisfied and evaluate the predictions. Compare with the predictions from logistic regression.

Solution:

```
rf_op <- randomForest(formula_op,data=training,na.action = na.omit)
pred_rf_op <- predict(rf_op,newdata=test)
conf_matrix <- table(test$satisfaction,pred_rf_op)
conf_matrix
```

```
##                pred_rf_op
##                 dissatisfied satisfied
##   dissatisfied          8128      3231
##   satisfied             2146     11495
```

```
prop.table(conf_matrix,margin=1)
```

```
##                pred_rf_op
##                 dissatisfied satisfied
##   dissatisfied    0.7155559 0.2844441
##   satisfied       0.1573198 0.8426802
```

```
overall_accuracy <- sum(diag(conf_matrix))/sum(conf_matrix)
overall_accuracy
```

```
## [1] 0.78492
```

Improved predictions compared with the logistic regression.

(l) (4 points) Consider all variables in the original dataset used in (a)-(g). Are there some of the information that is realistic to collect in the future which can help to improve the predictions. Assume that survey questions such as the ones in the original data set must be collected anonymously.

Solution:

Some of the ignored variables in the operational predictions like the customer's satisfaction on baggage handling and in-flight entertainment are difficult to envision to collect with anonymity. However some ignored variables, such as `Gate_location` might be possible to predict in advance, even if the gate is not entirely known a long time in advance of the flight.

References

Jana, Sayantan. n.d. "Airlines Customer Satisfaction. Data from Kaggle." https://www.kaggle.com/datasets/sjleshrac/airlines-customer-satisfaction.